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# Uriel Salazar

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## Education

**San Diego, CA** **University of California, San Diego** **Sept 2019 – Present**

- **Major**: Electrical Engineering: Machine Learning & Controls, B.S. (*expected June 2022*) (GPA: 3.38) (1st Gen)
- **Coursework**: AI Algorithms, GPU Coding, OOP, Data Analysis, Rapid Prototyping, Assembly, DSP, Circuit Design

## Work Experience

**Audio Captioning Programmer** **University of California, San Diego** **March 2021 – Sept 2021**

Project: *Sensing Everyday Activities Using Wi-Fi: A Machine Learning Based Framework* Mentor: Dr. Xinyu Zhang

- Utilized PyTorch to train a model that generates egocentric audio captioning based off user-POV GoPro video.
- Wrote a python script to extract over 700GB of audio from the EPIC KITHENS dataset via FFmpeg and Bash.
- Features were extracted from mp3 files to NumPy arrays (.npy) and connected to labels for writing the caption-generator model (.pt); the model was tested against the sample captions from the Clotho v2 dataset.

**Supplemental Instructor** **Riverside Academic Support** **July 2017 – June 2019**

- Restored underrepresented student graduation rates by composing weekly additional lectures in physics & computer science courses; utilized systems such as Mathematica to visualize and explore problems in detail.
- Restructured the dependency cycle between student and prof. by facilitating independent learning strategies.

**Image Processing Researcher** **University of California, Riverside** **June 2015 – Sept 2015**

Program Title: *Research Experiences in Integrated Computational Entomology* Investigator: Dr. Eamonn Keogh

- Detected Aedes Aegypti mosquitoes in a laboratory environment using computer vision, image segmentation, and histogram modification to aid in the prevention of the dengue virus in rural South America and Africa.
- Incorporated statistical data using R and MATLAB image processing to increase the efficiency of drone-delivered insecticides by predicting precise geographical locations of high mosquito activity/infestations.

## Projects

**Susumeru – Anime Recommender** [https://github.com/salazaru/Susumeru] **Nov 2020 – Present**

- Trained a model using a crawled MyAnimeList.net dataset of over 300,000 users and 80 million records.
- Utilized K-nearest neighbors and cosine similarity algorithms to find similarities between users and genres.
- Determined most popular series by country and age; analyzed genre popularity over time through user score.
- Cleaned and filtered distorted data; developed in a remote and collaborative 5-person team environment.

**SignVision – Sign Language Translator** [https://devpost.com/software/signvision] **January 2020**

Two-Way translator for sign language assistance using computer vision | *H.A.R.D. Hacks 2020* (24-hour hackathon)

- Incorporated *AlwaysAI* computer vision models to translate hand gesture to ASL output via neural network.
- 3D printed a fully-functioning robotic hand with finger movement driven via servo motors and Raspberry Pi3.
- A desired word can be input into the program and the robotic hand will physically demonstrate the sequence.

**Positive Edge – Line-tracking Car** [https://github.com/salazaru/plus-edge] **Sept 2019 – Dec 2019**

Microcontroller project based on reinforcement learning | 2nd Place UCSD Robotics Competition (seeding bracket)

- Utilized C++ and various Arduino libraries to input photosensor data into a proportion-integral-derivative (PID) error system that modified dyadic rear-wheel velocities depending on input data (brightness values.)
- Implemented software testing, debugging, and test-driven development around the Arduino Mega 2560 unit.

**yourTA Online – IoT Detection System** [https://devpost.com/software/yourta] **October 2019**

"Your Trash Assistant: Online" – *San Diego Hacks 2019* (48-hour hackathon) | Category Award Winner

- Developed a sensor-based recycling scoreboard concept for UC campuses by constructing a prototype circuit that included multiple sensors to measure the frequency and volume of proper recycling practices by campus.
- Developed a Node.js server to display updated sensor values in real-time via the ESP8266 Wi-Fi controller.
- Won the Northrop Grumman Award for "*Best IoT Device that Incorporates Multiple Nodes.*"

**Syrus – AR Navigation Assistant** [https://github.com/Stents-/SyrusSUITS] **Sept 2018 – April 2019**

Program: *Spacesuit User-Interface Technologies for Students (SUITS)* | NASA Design Challenge Winner

- Developed a navigation system that included pathfinding, virtual arrows, user interface, and a 3D mini-map in C# and Unity to train and assist astronauts aboard the International Space Station complete objectives.
- Programmed using Microsoft HoloLens framework; utilized Leap Motion Controller for optical hand-tracking.

## Skills

**Languages**: *experienced*: C/C++, Python, Java. *familiar*: MATLAB, HTML/CSS, C#, R, JavaScript, SQL, PHP, CUDA
**Technology**: PyTorch, NumPy, SciPy, Scikit-learn, Pandas, git, Linux/Unix, Unity, Node.js, OrCad Capture, PSpice
**Spoken Languages**: *fluent*: English, Spanish. *Beginner*: Japanese（日本語: 初心者）**Citizenship Status**: U.S. Citizen